

Encryption of payload for Interoperability

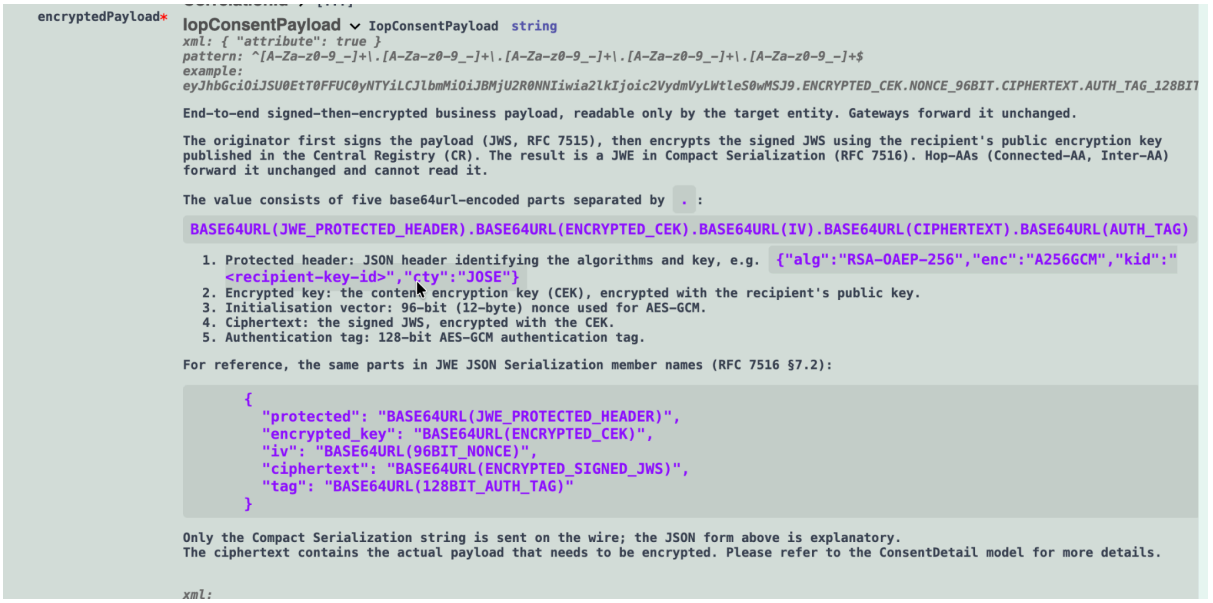
v1.0

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1.0	Jithesh Kumar		Initial document for encryption

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Purpose

This document explains what is the encryption involved in interoperability APIs



encryptedPayload carries the API's business payload end-to-end so that only the intended recipient can read it. The originator signs the payload (JWS) and then encrypts it using the recipient's public encryption key published in the Central Registry (CR). The result is a JWE in Compact Serialization (RFC 7516). Hops (Connected-AA, Inter-AA) forward it unchanged and cannot read it.

Illustrative flow considered

In this document, we'll look at a scenario where an FIU wants to make an /FI/request call to an AA that is not connected to it.

FIU: fiu-4 (the fiu where the customer has approached)

AA: aa-3 (the aa where the customer's handle and the consent reside)

Connectivity:

- fiu-4 has no connectivity to aa-3
- fiu-4 is contracted with aa-4

Available connectivity between fiu-4 and aa-3:

FIU(fiu-4) ⟶ Connected-AA(aa-4) ⟶ Customer-AA(aa-3)

The default payload

The default (direct) payload that the FIU would generate for an /FI/request call to AA as per the existing ReBIT specifications is as below:

```
Headers:
client_api_key = < Sahamati provided token for fiu-4 >
x-jws-signature = < fiu-4 generated signature of the below json >

Body:
{
  "ver": "2.0.0",
  "timestamp": "2026-10-07T07:35:00.000Z",
  "txnid": "e8cc6822-d4bb-4eb1-9e1b-4996fbff8acb",
  "FIDataRange": {
    "from": "2023-07-06T11:39:57.153Z",
    "to": "2019-12-06T11:39:57.153Z"
  },
  "Consent": {
    "id": "654024c8-29c8-11e8-8868-0289437bf331",
    "digitalSignature": "NdjwMjLortTb10dxcJezkvdOxPVvEdZvIqwq.....Ku78F69Z9Ig"
  },
  "KeyMaterial": {
    "cryptoAlg": "ECDH",
    "curve": "Curve25519",
    "params": "cipher=AES/GCM/NoPadding;KeyPairGenerator=ECDH",
    "DHPublicKey": {
      "expiry": "2023-07-06T11:39:57.153Z",
      "Parameters": "string",
      "KeyValue": "string"
    },
    "Nonce": "29512b70-ca84-46b5-9471-63765599cf15"
  }
}
```

Since fiu-4 doesn't have direct connectivity to aa-3, it decides to use aa-4 as a hop to get to aa-3, using the /iop/FI/request instead.

Plain-text before Encryption

Generate the plain-text to be encrypted.

Inputs:

a) Base64url encoded output of the payload: base64url encoded output of the direct ReBIT API payload.

```
ewogICJ2ZXIiOiAiMi4wLjAiLAogICJ0aW1lxNSIKICB9Cn0=
```

b) x-jws-signature, the detached jws signature of the payload: This contains JWS header, two dots, signature. The middle segment (payload segment) is empty. For example,

```
<JWS header>..<signature>
eyJhbGciOiJSU2LTEwIn0..rhAC_nMKavdp77CNAPYZvyV-f6bPssQAMKo1DaQEIOnhfZ
```

Plain-text for encryption is generated by inserting the base64url output of payload in between the two dots.

```
<JWS header> . <base64url(payload)> . <signature>
eyJhbGciOiJSU2LTEwIn0.ewogICJ2ZXIiOiAiMi4wLjAiLAogICJ0aW1lxNSIKICB9Cn0=.rhAC_nMKavdp7
7CNAPYZvyV-f6bPssQAMKo1DaQEIOnhfZ
```

This content is now encrypted. Refer to the sequence below and the sample code.

The Envelope payload

Generate a new envelope (or wrapper) request for the /iop/FI/request API, as below:

```
Headers:
client_api_key = < Sahamati provided token for fiu-4 >
x-jws-signature = < fiu-4 generated signature of the below wrapper json >

Body:
{
  "ver": "1.0.0",
  "timestamp": "2026-10-07T07:35:00.000Z",
  "txnid": "4a4adbbe-29ae-11e8-a8d7-0289437bf331",
  "correlationId": "550e8400-e29b-41d4-a716-446655440000",
  "Target": {
    "type": "AA",
    "id": "aa-3"
  },
  "encryptedPayload":
"<PROTECTED_HEADER>.<ENCRYPTED_KEY>.<IV>.<CIPHERTEXT>.<TAG>"
}
```

encryptedPayload field in the envelope is the signed and encrypted original request as per /FI/request call.

On the Connected-AA ⟶ Customer-AA hop, **Target** is replaced by **Initiator** (type, id of the originator) and encryptedPayload is passed through unchanged.

```
Headers:
x-conn_aa_api_key = < Sahamati provided token for aa-4 >
x-jws-signature = < aa-4 generated signature of the below wrapper json >

Body:
{
  "ver": "1.0.0",
  "timestamp": "2026-10-07T07:35:00.010Z",
  "txnid": "5a4adbbe-29ae-11e8-a8d7-0289437bf331",
  "correlationId": "550e8400-e29b-41d4-a716-446655440000",
  "Initiator": {
    "type": "FIU",
    "id": "fiu-4"
  },
  "encryptedPayload":
"<PROTECTED_HEADER>.<ENCRYPTED_KEY>.<IV>.<CIPHERTEXT>.<TAG>"
}
```

Note: The timestamp changes, the txnid changes. The connected-AA must retain the correlationId when forwarding the encryptedPayload. The encryptedPayload is forwarded as it is, i.e., it is still the signed and then encrypted original payload corresponding to /FI/request API.

Envelope payload for Error

Let's assume, for the FI/request call, the customer-AA returns an error, say Idempotency error, 409-Conflict.

The business payload in the response would be the following

```
Headers:
aa_api_key = < Sahamati provided token for aa-3 >
x-jws-signature = < aa-3 generated signature of the below wrapper json >

Body:
{
  "ver": "2.0.0",
  "txnid": "0b811819-9044-4856-b0ee-8c88035f8858'",
  "timestamp": "2023-06-26T11:33:34.509Z",
  "errorCode": "IdempotencyError",
  "errorMsg": "Error code specific error message"
}
```

However, since its an /iop/FI/request call, the above json payload would be signed and encrypted into encryptedPayload, as below

```
Headers:
x-cust_aa_api_key = < Sahamati provided token for aa-3 >
x-jws-signature = < aa-3 generated signature of the below wrapper json >

Body:
{
  "ver": "2.0.0",
  "txnid": "0b811819-9044-4856-b0ee-8c88035f8858",
  "timestamp": "2023-06-26T11:33:34.509Z",
  "correlationId": "550e8400-e29b-41d4-a716-446655440000",
  "encryptedPayload":
"eyJhbGciOiJSU0EtT0FFUC0yNTYiLCJlbmMiOiJBMjU2R0NNIiwia2lkIjoic2VydmVyLWtleS0wMSJ
9.ENCRYPTED_CEK.NONCE_96BIT.CIPHERTEXT.AUTH_TAG_128BIT"
}
```

The five parts of encryptedPayload

The parts appear in the order below, separated by "`.`", each base64url-encoded without padding (RFC 7516 7.1).

#	Part	Content	How it is produced	Size
1	Protected header	JSON object describing how the payload was encrypted	Written by the sender and base64url-encoded. The encoded value is also the AES-GCM additional authenticated data, so any change to it makes decryption fail.	Variable (≈100–150 characters)
2	Encrypted key	The content encryption key (CEK), wrapped with the recipient's public key.	A fresh random 256-bit CEK is generated for every message and wrapped with the recipient's encryption key from the CR, identified by kid.	256 bytes (RSA-2048); 512 bytes (RSA-4096)
3	Initialization vector (IV)	The AES-GCM nonce.	Fresh random value for every message; never reused with the same CEK.	12 bytes (96 bits)
4	Ciphertext	The encrypted plaintext. The plaintext is the originator's signed message (compact JWS).	AES-256-GCM encryption of the plaintext using the CEK and IV.	Same length as the plaintext
5	Authentication tag	The AES-GCM tag.	Output of the encryption step. The recipient checks it to detect any change to parts 1, 3 or 4.	16 bytes (128 bits)

Protected header parameters

Decoded form of Part-1:

```
{
  "alg": "RSA-OAEP-256",
  "enc": "A256GCM",
  "kid": "aa-003-enc-2026-10",
  "cty": "JWT"
}
```

Part	Content	How it is produced	Size
alg	Yes	RSA-OAEP-256	How the CEK is wrapped (part 2).
enc	Yes	A256GCM	How the plaintext is encrypted (parts 3–5).
kid	Yes	Recipient's encryption key id as published in the CR	Tells the recipient which private key to use; enables key rotation.
cty	Yes	JWT	The plaintext is a signed message that must be verified after decryption.

Sequence of generation of Encrypted field (sender)

1. Build the API's business payload.
2. Sign it with the originator's private signing key as a compact JWS. The JWS header carries alg and the signer's kid.
3. Look up the recipient's current encryption public key (for RSA-OAEP-256) and the kid in the CR (the Target.id on a request; the original sender on a response).
4. Generate a fresh random 256-bit CEK and a fresh random 96-bit IV.
5. Build the protected header (alg, enc, kid, cty) and base64url-encode it. This is part 1.
6. Wrap the CEK with the recipient's public key using RSA-OAEP-256. This is part 2.
7. Encrypt the JWS with AES-256-GCM using the CEK, the IV and the ASCII of part 1 as additional authenticated data. This yields the ciphertext (part 4) and tag (part 5).
8. Join parts 1–5, each base64url-encoded without padding, with . to form encryptedPayload.
9. Discard the CEK. Build the envelope (ver, timestamp, txnid, correlationId, routing block (Target), encryptedPayload), sign it for x-jws-signature, and send the API key in the authorization header as applicable for the interface.

Sequence of reading of Encrypted field (recipient)

1. Validate the hop: the caller's API key and the envelope's x-jws-signature against the caller's public key in the CR.
2. Split encryptedPayload on . and check that there are exactly five parts.
3. Decode the protected header; reject any alg or enc not on the allowed list.
4. Use kid to select the matching private key and unwrap the CEK with RSA-OAEP-256.
5. Decrypt with AES-256-GCM using the IV and the ASCII of part 1 as additional authenticated data. Reject if the tag check fails.
6. Verify the inner JWS against the signer's public key in the CR (selected by the JWS kid).
7. Check the signer: it must equal Initiator.id on a request, or the Target.id the request was sent to on a response. Reject otherwise.
8. Process the business payload.

Note

The same procedure applies, when an AA sends a request to FIP, or an FIP sends a notification to AA, or an AA sends a notification to FIU, which are not connected to them.